SSC MTS 2024 question paper with answers

29 October 2024 · Shift 3

78 questions as the Staff Selection Commission (SSC) printed them · answers from the official key · marking: 3 marks a question; no negative marking in Session-I (Numerical and Mathematical Ability, Reasoning), 1 mark deducted for each wrong answer in Session-II (General Awareness, English).

The other 12 of the paper's 90 questions are not reproduced: they need a figure or are printed as an image, or ask about formatting the source does not keep.

Try every question first: the answer key and the explanations follow the questions. Official answer-key notice: https://ssc.gov.in/api/attachment/uploads/masterData/AnswerKeys/approved_writeup_26425.pdf

Numerical and Mathematical Ability 12 questions

Q.1

What sum (in ₹) will earn a simple interest of ₹240 in 4 years at 16% per year rate of interest?

(A) 800
(B) 375
(C) 750
(D) 500

Q.2

Find the simple interest (in ₹) on ₹2500 as a sum borrowed at 4% per year rate of interest for 2 years.

(A) 160
(B) 240
(C) 140
(D) 200

Q.3

By selling an article for ₹20, a dealer makes a profit of 10%. By how much should he increase his selling price (in ₹, rounded off to two decimal place) so as to make a profit of 30%?

(A) 6.64
(B) 3.64
(C) 4.64
(D) 5.64

Q.6

The radii of the bases of two cylinders and their heights are in the ratio 4 : 3 and 9 : 8, respectively. Find the ratio of their curved surface areas.

(A) 4 : 5
(B) 2 : 3
(C) 3 : 2
(D) 5 : 6

Q.9

In the first 24 matches, a cricket player averages 60 runs. The overall average for all 32 matches is 80. Determine the average for the last 8 matches.

(A) 130
(B) 110
(C) 140
(D) 120

Q.11

In a circular race of 1 km, two persons X and Y start with the speed of 9 km/h and 18 km/h, respectively, at the same time from the same point. When will they meet for the first time at the starting point when running in the same direction?

(A) 240 sec
(B) 360 sec
(C) 180 sec
(D) 400 sec

Q.12

Two athletes X and Y are jogging on a circular path whose circumference is 500 m. Athlete X jogs at a speed of 18 km/h and Y jog at 9 km/h, jogging in opposite directions. The time at which they will meet for the first time is:

(A) 50 s
(B) 200 s
(C) 66.66 s
(D) 27.77 s

Q.13

The diameter of a hollow hemisphere is 30 cm. Water is filled in it up to its brim. When spherical lead shots each of radius 0.75 cm are dropped into the hemisphere, two-fifth of the water flows out. How many lead shots were dropped into the hemisphere?

(A) 1600
(B) 1380
(C) 2250
(D) 1575

Q.14

In a math exam, two students scored 95, six students scored 90, three students scored 80, and one student scored 50. What was the average score of the class?

(A) 85
(B) 78.75
(C) 82
(D) 96.25

Q.16

Arjun mixed three different oils to form a mixture costing ₹25 per litre. The ratio of the three varieties of oil was 1 : 5 : 2. What is the cost (per litre in ₹) of the second variety if the cost per litre of the first and third varieties was ₹20 and ₹30, respectively?

(A) 26
(B) 28
(C) 25
(D) 24

Q.19

A grocer wishes to sell a mixture of two varieties of rice worth ₹48 per kg. In what ratio must he mix the rice to reach this selling price when the cost of one variety of rice is ₹42 per kg and the other is ₹60 per kg?

(A) 1 : 2
(B) 2 : 3
(C) 3 : 2
(D) 2 : 1

Q.20

The money invested by Gopal at simple interest amounts to ₹16,150 in 3 years and to ₹16,540 in 4 years. What was the principal amount invested by him?

(A) ₹14,980
(B) ₹11,150
(C) ₹18,150
(D) ₹16,150

Reasoning Ability and Problem Solving 16 questions

Q.1

What will come in place of the question mark (?) in the following equation if '+' and '–' are interchanged and '×' and '÷' are interchanged? $24 \times 12 - 5 \div 2 + 7 = ?$

(A) 8
(B) 5
(C) 3
(D) 7

Q.2

A, B, C, D, E, F and G are sitting around a circular table facing the centre (but not necessarily in the same order) . Only C sits between G and E. E sits third to the left of F. D sits to the immediate left of F. B is not an immediate neighbour of E. Who sits second to the right of A?

(A) D
(B) F
(C) B
(D) C

Q.4

In a code language, 'ZHBN' is coded as ‘BEFI’ and 'LFWJ' is coded as ‘NCAE’. How will 'HVDK' be coded in that language?

(A) KTJD
(B) JSHF
(C) JTHD
(D) KSIF

Q.5

Six employees Dhiraj, Roshan, Satish, Pawan, Tarun and Niraj are sitting around a circular table, facing the centre (but not necessarily in the same order). Satish is sitting to the immediate right of Pawan. Niraj is sitting second to the left of Satish. Tarun is sitting to the immediate left of Niraj. Only Tarun is sitting between Niraj and Roshan. Who is sitting between Satish and Roshan?

(A) Niraj
(B) Dhiraj
(C) Tarun
(D) Pawan

Q.6

Select the set in which the numbers are related in the same way as are the numbers of the given sets. (NOTE: Operations should be performed on the whole numbers, without breaking down the numbers into their constituent digits. E.g. 13 – Operations on 13 such as adding/subtracting/multiplying to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed.) (38, 31, 47) (42, 35, 51)

(A) (29, 21, 39)
(B) (46, 40, 49)
(C) (33, 24, 30)
(D) (54, 47, 63)

Q.7

What should come in place of the question mark (?) in the given series? 46, 44, 40, 38, 34, 32, ?

(A) 24
(B) 30
(C) 26
(D) 28

Q.8

Read the given statements and conclusions carefully. Assuming that the information given in the statements is true, even if it appears to be at variance with commonly known facts, decide which of the given conclusions logically follow(s) from the statements. Statements: Some mugs are cups. No paper is a mug. Conclusions: (I) Some cups are mugs. (II) No paper is a cup.

(A) Only conclusion (II) follows
(B) Neither conclusion (I) nor (II) follows
(C) Both conclusions (I) and (II) follow
(D) Only conclusion (I) follows

Q.10

Read the given statements and conclusions carefully. Assuming that the information given in the statements is true, even if it appears to be at variance with commonly known facts, decide which of the given conclusions logically follow(s) from the statements. Statements: All puppies are dogs. All dogs are animals. Conclusions: I. All animals are puppies. II. Some dogs are not puppies.

(A) Only conclusion II follows
(B) Neither conclusion I nor II follows
(C) Both conclusions I and II follow
(D) Only conclusion I follows

Q.11

In a code language, 'JSCQ' is coded as ‘MNJH’ and 'BYTE' is coded as ‘ETAV’. How will 'SEWX' be coded in that language?

(A) VXDM
(B) VZDO
(C) WXEM
(D) WZEO

Q.13

10 is related to 6 following a certain logic. Following the same logic, 14 is related to 8. To which of the following is 24 related following the same logic?(NOTE: Operations should be performed on the whole numbers, without breaking down the numbersinto its constituent digits. E.g. 13 – Operations on 13 such as adding /subtracting /multiplying etc. to 13can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1and 3 is not allowed)

(A) 11
(B) 13
(C) 16
(D) 15

Q.14

VOAH is related to YPBK in a certain way based on the English alphabetical order. In the same way, RGDS is related to UHEV. To which of the following is IBXC related, following the same logic?

(A) LCYF
(B) MDYG
(C) NEZF
(D) KCZD

Q.16

In a certain code language, ‘i am infinite’ is written as ‘la ma ra’ and ‘universe is infinite’ is written as ‘ra ba ta’. How is ‘infinite’ written in the given code language?

(A) ba
(B) ra
(C) ma
(D) la

Q.17

In a certain code language, ‘BUSY’ is coded as ‘8642’ and ‘SIDE’ is coded as ‘7536’.What is the code for ‘S’ in the given code language?

(A) 6
(B) 7
(C) 4
(D) 5

Q.18

TVRM is related to VUPN in a certain way based on the English alphabetical order. In the same way, ROEP is related to TNCQ. To which of the following is KOCB related, following the same logic?

(A) NMAC
(B) MNBC
(C) MNAC
(D) NMBC

Q.19

Which of the following interchanges of signs would make the given equation correct? $25 - 63 \div 3 + 7 \times 5 = 11$

(A) ÷ and –
(B) + and ÷
(C) – and ×
(D) + and –

Q.20

What should come in place of the question mark (?) in the given series based on the English alphabetical order ? DEF, JKL, PQR, VWX, ?

(A) UWZ
(B) BCD
(C) ZBE
(D) YZB

General Awareness 25 questions

Q.1

Who built an elegant two-storey ‘ship palace’ between two reservoirs named Jahaaz Mahal?

(A) Sultan Alaud-Din Khalji
(B) Sultan Muhammad Bin Tughlaq
(C) Sultan Ghiyasuddin Khilji
(D) Sultan Ghiyasuddin Tughlaq

Q.2

According to Census of India 2011, in which of the following Union Territories was the sex ratio recorded at 1037 females per 1000 males?

(A) Puducherry
(B) Chandigarh
(C) Lakshadweep
(D) Andaman and Nicobar Islands

Q.3

Name the social reformer of Bengal who played crucial role in the enactment of the Hindu Widows' Remarriage Act 1856.

(A) Ishwarchandra Vidyasagar
(B) Rabindranath Tagore
(C) Swami Vivekananda
(D) Satyendra Nath Bose

Q.4

Chip, Condor, Duff and Driver are some of the terms used in _____.

(A) Swimming
(B) Chess
(C) Golf
(D) Horse Polo

Q.5

Teejan Bai is associated with which of the following dances?

(A) Bagurumba
(B) Jhumar
(C) Garba
(D) Pandvani

Q.6

Which of the following is NOT included in the national income?

(A) Income received by a household for the use of their land
(B) Income received by a household for the use of their entrepreneurial skill
(C) Income received by a person as a donation
(D) Income received by a person for the use of his capital

Q.7

M Balamuralikrishna was awarded Padma Vibhushan in 1991 for which of the following fields?

(A) Carnatic music
(B) Folk music
(C) Jazz music
(D) Hindustani music

Q.8

In 2023, the Government of India proposed 'Amrit Dharohar Scheme' to be implemented over the next _____ to encourage optimal use of wetlands and enhance bio-diversity, carbon stock, eco-tourism opportunities and income generation for local communities.

(A) Seven Years
(B) Three Years
(C) Five Years
(D) Two Years

Q.9

Jon Beel mela is held every year in which of the following states?

(A) Telangana
(B) Haryana
(C) Assam
(D) Arunachal Pradesh

Q.10
Which ruler succeeded Chandragupta Maurya in 297 BCE?

(A) Kanishka
(B) Bindusara
(C) Ashoka
(D) Kumaragupta

Q.11
Which of the following Acts was provided for the establishment of a federal court in India?

(A) The Indian Councils Act of 1909
(B) The Regulating Act of 1773
(C) The Government of India Act of 1935
(D) The Government of India Act of 1919

Q.12
Several biological processes are catalysed by enzymes. Digestive enzymes are present in which part of the cell?

(A) Ribosome
(B) Mitochondria
(C) Lysosomes
(D) Golgi complex

Q.13
In which of the following cities of India was the first successful textile mill established?

(A) Mumbai
(B) Kolkata
(C) Ludhiana
(D) Chennai

Q.14
The Thermohaline Circulation directly influences which of the following crucial environmental factors all over the world?

(A) The landforms
(B) The climate
(C) Natural resources
(D) Pollution in the air

Q.15
According to the Census of India 2011, which state has the highest sex ratio?

(A) Karnataka
(B) Kerala
(C) Tamil Nadu
(D) Andhra Pradesh

Q.16
In which of the following books has the author described the horrors of the holocaust and the Second World War?

(A) I know why the caged bird sings
(B) The Diary of a young girl
(C) Dreams from my father
(D) Long walk to freedom

Q.17
Who among the following was the Sultan of Delhi when Bahmani kingdom was founded in 1347 CE?

(A) Muhammad bin Tughlaq
(B) Bahlul Lodi
(C) Mubarak Shah Khalji
(D) Ghiyasuddin Tughlaq

Q.18
Which of the following is NOT true regarding the Swaran Singh Committee?

(A) The Swaran Singh Committee was set up in 1976.
(B) It suggested nine items as duties of the citizens.
(C) The committee was established during Indira Gandhi's Government
(D) The 42nd Constitutional Amendment Act was based on its recommendations.

Q.19
Which house is considered as lower house in State Legislature?

(A) Vidhan Parishad
(B) Rajya Sabha
(C) Vidhan Sabha
(D) Lok Sabha

Q.20
What is the name of the process of organic matter being broken down into simpler compounds by decomposers?

(A) Photosynthesis
(B) Fermentation
(C) Respiration
(D) Decomposition

Q.21
The Union Territory of Puducherry is located in which of the following river basins?

(A) Yamuna
(B) Kaveri
(C) Karishna
(D) Mahanadi

Q.22
Proposed in 2023-24, which section of the Income Tax Act of 1961 exempts senior citizens aged 75 years and above from filing income tax returns?

(A) Section 139A
(B) Section 194P
(C) Section 234C
(D) Section 80c

Q.23
Among the following Para athletes, which athlete is associated with the High Jump event?

(A) Avani Lekhara
(B) Mariyappan Thangavelu
(C) Deepa Malik
(D) Ajeet Singh Yadav

Q.24
0.25 g of an organic compound gave 0.13 g of water after complete combustion. The percentage composition of hydrogen is _____.

(A) 5.78%
(B) 11.11%
(C) 52.00%
(D) 4.58%

Q.25
Who among the following received the Sangeet Natak Akademi Amrit Award 2022 for Kuchipudi dance?

(A) Smita Rajeshbhai Shastri
(B) Kumkum Lal
(C) Makhan Chandra Borah
(D) Charan Girdhar Chand

English Language and Comprehension 25 questions

Q.1
Correct the underlined part of the sentence. They have a “few” juice in the jar.

(A) fewer
(B) least
(C) little
(D) lesser

Q.2
Select the word that has the same meaning (SYNONYM) as the underlined word in the given sentence. The town, where Haruki lived, stood in “imminent” danger of tidal waves.

(A) prominent
(B) immaculate
(C) impending
(D) eminent

Q.3

Identify the ANTONYM of the word in brackets. The athlete's (languid) performance disappointed the fans as well as coaches.

(A) Relaxed
(B) Lethargic
(C) Listless
(D) Vigorous

Q.4

Select the most appropriate set of words to fill the blanks. I was trying to _____ a pen to sign a form as my brother had been _____ Rupees 500 for breaking the traffic rules.

(A) fined; find
(B) find; find
(C) find; fined
(D) fined; fined

Q.5

Select the most appropriate option that can substitute the underlined segment in the given sentence. This is the “old” tree in the colony.

(A) elder
(B) oldest
(C) most old
(D) older

Q.6

Select the most appropriate option that can substitute the underlined words in the given sentence. Raman is as “worse” as Raju in solving mathematical problems.

(A) bad
(B) best
(C) worst
(D) worsen

Q.7

Select the appropriate option that can substitute the underlined word in the following sentence to make it grammatically correct. This novel is “more interesting” of all in the library.

(A) most interesting
(B) interesting
(C) the most interesting
(D) the more interesting

Q.8

Select the option that can be used as one-word substitute for the given group of words A policy that separates people on the basis of race

(A) Apartheid
(B) Apostasy
(C) Apostle
(D) Apiary

Q.9

Rectify the sentence by selecting the correct spelling from the options. Despite his busy scheuduel, Rohit managed to complete the assignment on time.

(A) scheudul
(B) schedeule
(C) schedule
(D) scedhule

Q.10

Select the most appropriate option to fill in the blank. The arrangements of the little boy’s birthday party were _____ nice. We could not help but appreciate it.

(A) extremely
(B) little
(C) many
(D) few

Q.11

Select the most appropriate ANTONYM of the underlined word in the given sentence. Abhay's “humility” towards the poor and needy people of his locality is really admirable.

(A) Arrogance
(B) Modesty
(C) Amicability
(D) Sincerity

Q.12

Select the most appropriate meaning of the given phrase. Feel the pinch

(A) Raise quality
(B) Give pain
(C) Demand something
(D) Experience hardship

Q.13

Select the INCORRECTLY spelt word.

(A) Transcendental
(B) Fundementel
(C) Detrimental
(D) Regimental

Q.14

Select the most appropriate synonym of the given word. Ephemeral

(A) Durable
(B) Fleeting
(C) persistent
(D) Permanent

Q.15

Select the most appropriate meaning of the idiom from the options given below. A square peg in a round hole

(A) a person who does not fit the job
(B) a person who is bankrupt
(C) a person who fits the job
(D) a person who is criminal by nature

Q.16

Select the most appropriate option that can substitute the underlined segment in the given sentence. If there is no need to substitute it, select ‘No substitution’. While I was teaching the elder one, “the younger one take an opportunity and ran away”.

(A) the younger one has taken an opportunity and ran away
(B) the younger one took an opportunity and ran away
(C) No substitution
(D) the younger one will take an opportunity and ran away

Q.17

In the following sentence the underlined part contains an error. Alternatives to the underlined part are given as options. Select the correct alternative. The house seemed much “far” than she expected.

(A) farthest
(B) farther
(C) more far
(D) farer

Q.18

Select the option that can be used as a one-word substitute for the given group of words. The method and practice of teaching

(A) Objective
(B) Strategy
(C) Pedagogy
(D) Learning

Q.19

Select the most appropriate option that can substitute the underlined segment in the given sentence. This is the “cheap” saree in my collection.

(A) cheaper
(B) most cheapest
(C) cheapest
(D) more cheap

Q.20

Select the INCORRECTLY spelt word..

(A) Embarrassment
(B) Manoevre
(C) Consistent
(D) Occurrence

Q.21

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. There lived in a town in Russia a bootmaker named Martin. He lived in a small room, half sunk _____1_____ the level of the street, with one window which _____2_____ on the street. He had lived in the same room for many years and knew many people. There was _____3_____ a pair of boots in the town and in the country for many miles _____4_____ that had not once or twice passed _____5_____ his hands.

Select the most appropriate option to fill in blank number 1.

(A) beyond
(B) amidst
(C) below
(D) inside

Q.22

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. There lived in a town in Russia a bootmaker named Martin. He lived in a small room, half sunk _____1_____ the level of the street, with one window which _____2_____ on the street. He had lived in the same room for many years and knew many people. There was _____3_____ a pair of boots in the town and in the country for many miles _____4_____ that had not once or twice passed _____5_____ his hands.

Select the most appropriate option to fill in blank number 2.

(A) aimed
(B) touched
(C) opened
(D) arrived

Q.23

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. There lived in a town in Russia a bootmaker named Martin. He lived in a small room, half sunk _____1_____ the level of the street, with one window which _____2_____ on the street. He had lived in the same room for many years and knew many people. There was _____3_____ a pair of boots in the town and in the country for many miles _____4_____ that had not once or twice passed _____5_____ his hands.

Select the most appropriate option to fill in blank number 3.

(A) now
(B) hardly
(C) always
(D) besides

Q.24

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. There lived in a town in Russia a bootmaker named Martin. He lived in a small room, half sunk _____1_____ the level of the street, with one window which _____2_____ on the street. He had lived in the same room for many years and knew many people. There was _____3_____ a pair of boots in the town and in the country for many miles _____4_____ that had not once or twice passed _____5_____ his hands.

Select the most appropriate option to fill in blank number 4.

(A) above
(B) below
(C) around
(D) round

Q.25

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. There lived in a town in Russia a bootmaker named Martin. He lived in a small room, half sunk _____1_____ the level of the street, with one window which _____2_____ on the street. He had lived in the same room for many years and knew many people. There was _____3_____ a pair of boots in the town and in the country for many miles _____4_____ that had not once or twice passed _____5_____ his hands.

Select the most appropriate option to fill in blank number 5.

(A) in

(B) through

(C) off

(D) at

Answer key

Numerical and Mathematical Ability

Q.1 **(B)**	Q.2 **(D)**	Q.3 **(B)**	Q.6 **(C)**	Q.9 **(C)**	Q.11 **(D)**
Q.12 **(C)**	Q.13 **(A)**	Q.14 **(A)**	Q.16 **(D)**	Q.19 **(D)**	Q.20 **(A)**

Reasoning Ability and Problem Solving

Q.1 **(B)**	Q.2 **(B)**	Q.4 **(B)**	Q.5 **(B)**	Q.6 **(D)**	Q.7 **(D)**
Q.8 **(D)**	Q.10 **(B)**	Q.11 **(B)**	Q.13 **(B)**	Q.14 **(A)**	Q.16 **(B)**
Q.17 **(A)**	Q.18 **(C)**	Q.19 **(D)**	Q.20 **(B)**		

General Awareness

Q.1 **(C)**	Q.2 **(A)**	Q.3 **(A)**	Q.4 **(C)**	Q.5 **(D)**	Q.6 **(C)**
Q.7 **(A)**	Q.8 **(B)**	Q.9 **(C)**	Q.10 **(B)**	Q.11 **(C)**	Q.12 **(C)**
Q.13 **(A)**	Q.14 **(B)**	Q.15 **(B)**	Q.16 **(B)**	Q.17 **(A)**	Q.18 **(B)**
Q.19 **(C)**	Q.20 **(D)**	Q.21 **(B)**	Q.22 **(B)**	Q.23 **(B)**	Q.24 **(A)**
Q.25 **(A)**					

English Language and Comprehension

Q.1 **(C)**	Q.2 **(C)**	Q.3 **(D)**	Q.4 **(C)**	Q.5 **(B)**	Q.6 **(A)**
Q.7 **(C)**	Q.8 **(A)**	Q.9 **(C)**	Q.10 **(A)**	Q.11 **(A)**	Q.12 **(D)**
Q.13 **(B)**	Q.14 **(B)**	Q.15 **(A)**	Q.16 **(B)**	Q.17 **(B)**	Q.18 **(C)**
Q.19 **(C)**	Q.20 **(B)**	Q.21 **(C)**	Q.22 **(C)**	Q.23 **(B)**	Q.24 **(C)**
Q.25 **(B)**					

Explanations

Numerical and Mathematical Ability

Q.1 · Answer (B) 375

P = SI × 100/(R × T) = 240 × 100/(16 × 4) = ₹375.

Q.2 · Answer (D) 200

SI = 2,500 × 4 × 2/100 = ₹200.

Q.3 · Answer (B) 3.64

Cost price = 20/1.1 = ₹200/11. At 30% profit the price must be 1.3 × 200/11 = ₹260/11 ≈ ₹23.64, an increase of 40/11 ≈ ₹3.64.

Q.6 · Answer (C) 3 : 2

Curved surface area = 2πrh, so the ratio is (4 × 9) : (3 × 8) = 36 : 24 = 3 : 2.

Q.9 · Answer (C) 140

All 32 matches: 32 × 80 = 2,560 runs; the first 24: 24 × 60 = 1,440. The last 8 made 1,120 runs, an average of 140.

Q.11 · Answer (D) 400 sec

X takes 1/9 h = 400 s to run a lap and Y 1/18 h = 200 s. They are next together at the start after LCM(400, 200) = 400 s.

Q.12 · Answer (C) 66.66 s

18 km/h = 5 m/s and 9 km/h = 2.5 m/s. Running in opposite directions they close the 500 m at 7.5 m/s, so they meet after 500/7.5 = 66.66... s.

Q.13 · Answer (A) 1600

With radius 15 cm the hemisphere holds (2/3)π × 15^3 = 2,250π cm^3, so two-fifths, 900π cm^3, flows out. Each shot is (4/3)π × 0.75^3 = 0.5625π cm^3: 900/0.5625 = 1,600 shots.

Q.14 · Answer (A) 85

(2 × 95 + 6 × 90 + 3 × 80 + 50)/12 = (190 + 540 + 240 + 50)/12 = 1,020/12 = 85.

Q.16 · Answer (D) 24

(1 × 20 + 5x + 2 × 30)/8 = 25, so 80 + 5x = 200 and x = ₹24.

Q.19 · Answer (D) 2 : 1

By alligation, cheaper : dearer = (60 − 48) : (48 − 42) = 12 : 6 = 2 : 1.

Q.20 · Answer (A) ₹14,980

One year's interest is 16,540 − 16,150 = ₹390, so three years' is ₹1,170 and the principal ₹16,150 − 1,170 = ₹14,980.

Reasoning Ability and Problem Solving

Q.1 · Answer (B) 5

After interchanging the signs: 24 ÷ 12 + 5 × 2 − 7 = 2 + 10 − 7 = 5.

Q.2 · Answer (B) F

Facing the centre, each person's left is clockwise. Clockwise from F come D, then a seat, then E; with only C between G and E the order goes on E, C, G. B cannot sit next to E, so B is on F's right and A between D and E: F, D, A, E, C, G, B. Second to A's right is F.

Q.4 · Answer (B) JSHF

The letters move +2, −3, +4, −5: ZHBN → BEFI and LFWJ → NCAE. So HVDK → JSHF.

Q.5 · Answer (B) Dhiraj
Going to each person's left from Pawan: Niraj, Tarun, Roshan, Dhiraj, Satish. Satish is on Pawan's right, Niraj second to Satish's left, Tarun on Niraj's left and Roshan just beyond Tarun; Dhiraj takes the last seat, between Roshan and Satish.

Q.6 · Answer (D) (54, 47, 63)
In each set the second number is the first minus 7 and the third the first plus 9: 38 – 7 = 31, 38 + 9 = 47; 42 – 7 = 35, 42 + 9 = 51. Likewise 54 – 7 = 47 and 54 + 9 = 63.

Q.7 · Answer (D) 28
The differences alternate –2 and –4: 46, 44, 40, 38, 34, 32, so next 32 – 4 = 28.

Q.8 · Answer (D) Only conclusion (I) follows
'Some mugs are cups' converts to 'some cups are mugs' (I). Paper is only kept out of mugs; cups that are not mugs may still be paper, so II does not follow.

Q.10 · Answer (B) Neither conclusion I nor II follows
Not every animal need be a puppy, so I does not follow; and every dog could be a puppy, so II need not follow either.

Q.11 · Answer (B) VZDO
The letters move +3, –5, +7, –9, wrapping round the alphabet: JSCQ → MNJH and BYTE → ETAV. So SEWX → VZDO.

Q.13 · Answer (B) 13
Half the number plus 1: 10/2 + 1 = 6 and 14/2 + 1 = 8. So 24/2 + 1 = 13.

Q.14 · Answer (A) LCYF
The letters move +3, +1, +1, +3: VOAH → YPBK and RGDS → UHEV. So IBXC → LCYF.

Q.16 · Answer (B) ra
'infinite' is the only word in both sentences and 'ra' the only code in both, so infinite = ra.

Q.17 · Answer (A) 6
S is the only letter in both words and 6 the only digit in both codes, so S = 6.

Q.18 · Answer (C) MNAC
The letters move +2, –1, –2, +1: TVRM → VUPN and ROEP → TNCQ. So KOCB → MNAC.

Q.19 · Answer (D) + and –
Interchanging + and –: 25 + 63 ÷ 3 – 7 × 5 = 25 + 21 – 35 = 11.

Q.20 · Answer (B) BCD
Each group moves 6 places forward: D, J, P, V, and V + 6 wraps round to B. So BCD.

General Awareness

Q.1 · Answer (C) Sultan Ghiyasuddin Khilji
The Jahaz Mahal at Mandu, between the Munj and Kapur lakes, was built by Ghiyasuddin Khilji of the Malwa Sultanate in the late 15th century.

Q.2 · Answer (A) Puducherry
Census 2011 recorded 1,037 females per 1,000 males in Puducherry, the highest of any Union Territory.

Q.3 · Answer (A) Ishwarchandra Vidyasagar
Ishwarchandra Vidyasagar campaigned for widow remarriage, and his petitions led to the Hindu Widows' Remarriage Act of 1856.

Q.4 · Answer (C) Golf
These are golf terms: a chip is a short lofted shot, a condor is four strokes under par on a hole, a duff is a badly mishit shot and a driver is the club for long tee shots.

Q.5 · Answer (D) Pandvani
Teejan Bai is the best-known exponent of Pandavani, Chhattisgarh's folk art of narrating and acting out the Mahabharata in song.

Q.6 · Answer (C) Income received by a person as a donation
A donation is a transfer payment, not a return for any productive service, so it is not part of national income. Rent (land), profit (enterprise) and interest (capital) are factor incomes and are counted.

Q.7 · Answer (A) Carnatic music
M. Balamuralikrishna was a celebrated Carnatic vocalist and composer.

Q.8 · Answer (B) Three Years
The Union Budget 2023-24, presented on 1 February 2023, said the Amrit Dharohar scheme would be implemented over the next three years.

Q.9 · Answer (C) Assam
The Jonbeel Mela, held in Assam's Morigaon district, is known for the barter trade between the Tiwa and other communities.

Q.10 · Answer (B) Bindusara
Chandragupta Maurya was succeeded by his son Bindusara, the father of Ashoka.

Q.11 · Answer (C) The Government of India Act of 1935
The Government of India Act, 1935 provided for a Federal Court, which was set up in 1937 and was the forerunner of the Supreme Court.

Q.12 · Answer (C) Lysosomes
Lysosomes are membrane-bound sacs of digestive (hydrolytic) enzymes that break down worn-out cell parts and foreign material.

Q.13 · Answer (A) Mumbai
The first successful textile mill in India was set up in Mumbai in 1854.

Q.14 · Answer (B) The climate
The thermohaline circulation, driven by differences in the temperature and salinity of sea water, carries heat round the world's oceans and so shapes the climate everywhere.

Q.15 · Answer (B) Kerala
Census 2011 recorded Kerala's sex ratio as 1,084 females per 1,000 males, the highest of any state.

Q.16 · Answer (B) The Diary of a young girl
'The Diary of a Young Girl' is Anne Frank's diary, kept while her Jewish family hid from the Nazis in occupied Amsterdam.

Q.17 · Answer (A) Muhammad bin Tughlaq
Alauddin Hasan Bahman Shah founded the Bahmani kingdom in 1347 after the Deccan revolted against Muhammad bin Tughlaq, Sultan from 1325 to 1351.

Q.18 · Answer (B) It suggested nine items as duties of the citizens.
The Swaran Singh Committee, set up in 1976, recommended eight Fundamental Duties, not nine; the 42nd Amendment of 1976 added ten.

Q.19 · Answer (C) Vidhan Sabha
The Vidhan Sabha (Legislative Assembly) is a state's lower house and the Vidhan Parishad (Legislative Council) its upper house; the Lok Sabha and Rajya Sabha are the houses of Parliament.

Q.20 · Answer (D) Decomposition
In decomposition, decomposers such as bacteria and fungi break dead organic matter down into simpler substances.

Q.21 · Answer (B) Kaveri
Of the four, only the Kaveri basin takes in part of Puducherry: the territory's Karaikal district lies in the Kaveri delta.

Q.22 · Answer (B) Section 194P
Under Section 194P a specified bank deducts the tax of a resident senior citizen aged 75 or more whose only income is pension and interest from that bank, who then need not file a return. (The section came in with the Finance Act 2021, not in 2023-24.)

Q.23 · Answer (B) Mariyappan Thangavelu
Mariyappan Thangavelu is a Paralympic high jumper, gold medallist at Rio 2016. Avani Lekhara is a shooter, Deepa Malik a shot putter and Ajeet Singh Yadav a javelin thrower.

Q.24 · Answer (A) 5.78%
Hydrogen is 2/18 of the mass of water, so % H = (2/18) × (0.13/0.25) × 100 = 52/9 ≈ 5.78%.

Q.25 · Answer (A) Smita Rajeshbhai Shastri
Smita Shastri, a Kuchipudi dancer from Gujarat, was among the artists over 75 given the one-time Amrit Award of 2022, marking 75 years of independence. Kumkum Lal is an Odissi dancer.

English Language and Comprehension

Q.1 · Answer (C) little
'Juice' is uncountable, so it takes 'a little'; 'a few' goes with plural countable nouns.

Q.2 · Answer (C) impending
'Imminent' means about to happen: 'impending'. 'Eminent' (distinguished) only looks similar.

Q.3 · Answer (D) Vigorous
'Languid' means slow and lacking energy; its opposite is 'vigorous'. The other three are its synonyms.

Q.4 · Answer (C) find; fined
One tries to 'find' a pen; the brother had been 'fined', made to pay a penalty, for breaking the rules.

Q.5 · Answer (B) oldest
Comparing one tree with all the others in the colony needs the superlative: 'the oldest tree'.

Q.6 · Answer (A) bad
'As … as' takes the positive degree: 'as bad as'.

Q.7 · Answer (C) the most interesting
'Of all' calls for the superlative with 'the': 'the most interesting of all'.

Q.8 · Answer (A) Apartheid
'Apartheid' is a policy of separating people by race, as in South Africa from 1948 to the early 1990s.

Q.9 · Answer (C) schedule
The correct spelling is 'schedule'.

Q.10 · Answer (A) extremely
Only an adverb of degree can qualify the adjective 'nice': 'extremely nice'.

Q.11 · Answer (A) Arrogance
'Humility' is a modest view of oneself; its opposite is 'arrogance'. 'Modesty' is its synonym.

Q.12 · Answer (D) Experience hardship
To 'feel the pinch' is to suffer hardship, especially from a shortage of money.

Q.13 · Answer (B) Fundementel
The correct spelling is 'fundamental'.

Q.14 · Answer (B) Fleeting
'Ephemeral' means lasting a very short time: 'fleeting'. 'Durable', 'persistent' and 'permanent' are its opposites.

Q.15 · Answer (A) a person who does not fit the job
'A square peg in a round hole' is a person unsuited to their job or surroundings.

Q.16 · Answer (B) the younger one took an opportunity and ran away
The other verbs are in the past ('was teaching', 'ran'), so 'take' must be 'took'.

Q.17 · Answer (B) farther
A comparison with 'than' needs the comparative: 'much farther than'.

Q.18 · Answer (C) Pedagogy
'Pedagogy' is the method and practice of teaching; 'learning' is what the pupil does, and a 'strategy' is any plan of action.

Q.19 · Answer (C) cheapest
Comparing one saree with all the others in the collection needs the superlative: 'the cheapest'.

Q.20 · Answer (B) Manoevre
The correct spelling is 'manoeuvre' (American 'maneuver').

Q.21 · Answer (C) below
The room was 'half sunk below the level of the street': partly underground.

Q.22 · Answer (C) opened
A window that 'opened on' the street looked out on it.

Q.23 · Answer (B) hardly
'There was hardly a pair of boots … that had not' passed through his hands: nearly every pair had.

Q.24 · Answer (C) around
SSC's key gives 'around': 'for many miles around' means in all the surrounding country. ('Round' is used the same way in British English, as in 'for miles round', so it is also arguable.)

Q.25 · Answer (B) through
Boots that 'passed through his hands' were ones he had worked on.

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